Section 7.1: Sets

Finite Mathematics MTH 132 Spring 2014

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Chapter 7: Sets and Probability

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7.1: Sets

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7.2: Venn Diagrams

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7.1: Sets
7.2: Venn Diagrams
7.3: Introduction to Probability

ex. $\{0, 2, 4\}$

Set

A *set* is a well defined collection of distinct objects.

ex. $\{0, 2, 4\}$

ex. $N = \{Billy, Bob, Bubba\}$

Sets

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- Usually written as capital letters, e.g., A, B, C, etc.

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ex. $N = \{Billy, Bob, Bubba\}$

ex. The set of letters in the alphabet

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ex. The set of positive integers.

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$$\{a,b,c,\ldots,x,y,z\}$$

ex. The set of positive integers.

$$\{1,2,3,4,\ldots\}$$

ex. The set of positive integers< 0.

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$$\emptyset \ (\emptyset \neq 0)$$

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Element

The objects in the set are called *elements* or *members*.

- $x \in A$ means x is an element in A.
- $x \notin A$ means x is not an element in A.

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ex. $2 \in \{0, 2, 4\}$

Sets

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ex. $2 \in \{0, 2, 4\}$

ex. $Carlton \notin N$

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ex. $Carlton \notin N$

ex. $h \in \{a,b,c,\ldots,x,y,z\}$

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The objects in the set are called *elements* or *members*.

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ex. $2 \in \{0,2,4\}$

ex. $Carlton \notin N$

ex. $h \in \{a,b,c,\ldots,x,y,z\}$

ex. $3.14 \notin \{1,2,3,4,\ldots\}$

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ex. The set of positive integers.

$$\{1, 2, 3, 4, \ldots\}$$

ex. The set of positive integers< 0.

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Element

The objects in the set are called *elements* or *members*.

- $x \in A$ means x is an element in A.
- $x \notin A$ means x is not an element in A.

ex. $2 \in \{0, 2, 4\}$

ex. $Carlton \notin N$

ex. $h \in \{a, b, c, \ldots, x, y, z\}$

ex. $3.14 \notin \{1, 2, 3, 4, \ldots\}$

ex. $x \notin \emptyset$

Sets

- Two sets are *equal* if the contain the same elements.

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.

ex. $\{0,2,4\} \neq \{0,2,4,1\}$

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ex. $\{0,2,4\} \neq \{0,2,4,1\}$

ex. $\{0,2,4\} = \{2,4,0\} = \{4,0,2\}$

Sets

- Two sets are *equal* if the contain the same elements.
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    - We do not repeat elements

ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

ex. $\{0, 2, 4\} = \{2, 4, 0\} = \{4, 0, 2\}$

Sets

- Two sets are *equal* if the contain the same elements.
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ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

ex. $\{0, 2, 4\} = \{2, 4, 0\} = \{4, 0, 2\}$

Do not write $\{0, 2, 2, 4\}$

Sets

- Two sets are *equal* if the contain the same elements.
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A set can be given by,

ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

ex. $\{0, 2, 4\} = \{2, 4, 0\} = \{4, 0, 2\}$

Do not write $\{0, 2, 2, 4\}$

Sets

- Two sets are *equal* if the contain the same elements.
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A set can be given by,

1. description

ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

ex. $\{0, 2, 4\} = \{2, 4, 0\} = \{4, 0, 2\}$

Do not write $\{0, 2, 2, 4\}$

- Two sets are *equal* if the contain the same elements.
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ex. $\{0,2,4\} \neq \{0,2,4,1\}$

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Do not write $\{0,2,2,4\}$

A set can be given by,

1. description

ex. The set of positive integers

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.
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A set can be given by,

1. description
2. *tabular* form

ex. $\{0,2,4\} \neq \{0,2,4,1\}$

ex. $\{0,2,4\} = \{2,4,0\} = \{4,0,2\}$

Do not write $\{0,2,2,4\}$

ex. The set of positive integers

Sets

- Two sets are *equal* if the contain the same elements.
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A set can be given by,

1. description
2. *tabular* form

ex. The set of positive integers

ex. $\{1,2,3,4,\dots\}$

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.
  - We do not repeat elements

ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

ex. $\{0, 2, 4\} = \{2, 4, 0\} = \{4, 0, 2\}$

Do not write $\{0, 2, 2, 4\}$

A set can be given by,

1. description
2. *tabular* form
3. *set-builder* notation

ex. The set of positive integers

ex. $\{1, 2, 3, 4, \dots\}$

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.
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ex. $\{0,2,4\} \neq \{0,2,4,1\}$

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Do not write $\{0,2,2,4\}$

A set can be given by,

1. description
2. *tabular* form
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ex. The set of positive integers

ex. $\{1,2,3,4,\dots\}$

ex. $\{x \mid x > 0 \text{ and } x \text{ is an integer}\}$

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.
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ex. $\{0, 2, 4\} \neq \{0, 2, 4, 1\}$

ex. $\{0, 2, 4\} = \{2, 4, 0\} = \{4, 0, 2\}$

Do not write $\{0, 2, 2, 4\}$

A set can be given by,

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ex. The set of positive integers

ex. $\{1, 2, 3, 4, \dots\}$

ex. $\{x \mid x > 0 \text{ and } x \text{ is an integer}\}$

Problem 7.1.1. Write the following set in *tabular* and *set-builder* form.

The set of even, positive integers.

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.
  - We do not repeat elements

ex. $\{0,2,4\} \neq \{0,2,4,1\}$

ex. $\{0,2,4\} = \{2,4,0\} = \{4,0,2\}$

Do not write $\{0,2,2,4\}$

A set can be given by,

1. description
2. *tabular* form
3. *set-builder* notation

ex. The set of positive integers

ex. $\{1,2,3,4,\dots\}$

ex. $\{x \mid x > 0 \text{ and } x \text{ is an integer}\}$

Problem 7.1.2. Decide whether the following are *true* or *false*.

$\{1,2,3,4\} = \{4,3,1,2\}$

Sets

- Two sets are *equal* if the contain the same elements.
  - Order does not matter.
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ex. $\{0,2,4\} \neq \{0,2,4,1\}$

ex. $\{0,2,4\} = \{2,4,0\} = \{4,0,2\}$

Do not write $\{0,2,2,4\}$

A set can be given by,

1. description
2. *tabular* form
3. *set-builder* notation

ex. The set of positive integers

ex. $\{1,2,3,4,\dots\}$

ex. $\{x \mid x > 0 \text{ and } x \text{ is an integer}\}$

Problem 7.1.3. Decide whether the following are *true* or *false*.

$$\{a,b,\dots,z\} = \{b,c,\dots,z\}$$

Subsets

Subsets

Subset

a set A is a *subset* of a set B if every element in A is also in B.

$A \subseteq B$: A is a subset of B

$A \not\subseteq B$: A is not a subset of B

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a set A is a *subset* of a set B if every element in A is also in B.

$A \subseteq B$: A is a subset of B

$A \not\subseteq B$: A is <u>not</u> a subset of B

ex.

$$A = \{3, 4, 5, 6\}$$
$$B = \{2, 3, 4, 5, 6, 7, 8\}$$
$$C = \{3, 4, 5, 6, 9\}$$

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a set A is a *subset* of a set B if every element in A is also in B.

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ex.
$$\begin{aligned} A &= \{3,4,5,6\} \\ B &= \{2,3,4,5,6,7,8\} \\ C &= \{3,4,5,6,9\} \end{aligned}$$

- $A \subseteq B$

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- $A \subseteq B$
- $C \not\subseteq B$

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- $A \subseteq B$
- $C \not\subseteq B$
- $\emptyset \subseteq A$, $\emptyset \subseteq B$, and $\emptyset \subseteq C$

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- $A \subseteq B$
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Subsets

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a set A is a *subset* of a set B if every element in A is also in B.

$A \subseteq B$: A is a subset of B

$A \not\subseteq B$: A is <u>not</u> a subset of B

Proper Subset

a set A is a *proper subset* of a set B if every element in A is also in B, and $A \neq B$.

$A \subset B$: A is a proper subset of B.

ex.
$$A = \{3,4,5,6\}$$
$$B = \{2,3,4,5,6,7,8\}$$
$$C = \{3,4,5,6,9\}$$

- $A \subseteq B$
- $C \not\subseteq B$
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- $B \subseteq B$, $C \subseteq C$, $A \subseteq A$

Subsets

Subset

a set A is a *subset* of a set B if every element in A is also in B.

$A \subseteq B$: A is a subset of B

$A \nsubseteq B$: A is <u>not</u> a subset of B

ex.
$$A = \{3, 4, 5, 6\}$$
$$B = \{2, 3, 4, 5, 6, 7, 8\}$$
$$C = \{3, 4, 5, 6, 9\}$$

- $A \subseteq B$
- $C \nsubseteq B$
- $\emptyset \subseteq A$, $\emptyset \subseteq B$, and $\emptyset \subseteq C$
- $B \subseteq B$, $C \subseteq C$, $A \subseteq A$

Proper Subset

a set A is a *proper subset* of a set B if every element in A is also in B, and $A \neq B$.

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Subset

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$A \subseteq B$: A is a subset of B

$A \not\subseteq B$: A is not a subset of B

ex.
$$A = \{3, 4, 5, 6\}$$
$$B = \{2, 3, 4, 5, 6, 7, 8\}$$
$$C = \{3, 4, 5, 6, 9\}$$

- $A \subseteq B$
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- $\emptyset \subseteq A$, $\emptyset \subseteq B$, and $\emptyset \subseteq C$
- $B \subseteq B$, $C \subseteq C$, $A \subseteq A$

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$$A = \{3, 4, 5, 6\}$$
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- $A \subset B$

Subsets

Subset

a set A is a *subset* of a set B if every element in A is also in B.

$A \subseteq B$: A is a subset of B

$A \not\subseteq B$: A is <u>not</u> a subset of B

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$$A = \{3, 4, 5, 6\}$$
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- $A \subseteq B$
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- $\emptyset \subseteq A$, $\emptyset \subseteq B$, and $\emptyset \subseteq C$
- $B \subseteq B$, $C \subseteq C$, $A \subseteq A$

Proper Subset

a set A is a *proper subset* of a set B if every element in A is also in B, and $A \neq B$.

$A \subset B$: A is a proper subset of B.

ex.
$$A = \{3, 4, 5, 6\}$$
$$B = \{2, 3, 4, 5, 6, 7, 8\}$$
$$C = \{3, 4, 5, 6, 9\}$$

- $A \subset B$
- B is not a proper subset of B.

Subsets

- $A \subseteq A$ for any set A.

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$

ex. $\{1, 2\}$ has 2^2 subsets.

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Problem 7.1.4. Let $A = \{2,4,6,10\}$, $B = \{2,10\}$, and $C = \{2,4,6,10\}$. Decide whether the following are true or false.

$B \subset A$

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Problem 7.1.5. Let $A = \{2,4,6,10\}$, $B = \{2,10\}$, and $C = \{2,4,6,10\}$. Decide whether the following are true or false.

$A \subseteq B$

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Problem 7.1.6. Let $A = \{2,4,6,10\}$, $B = \{2,10\}$, and $C = \{2,4,6,10\}$. Decide whether the following are true or false.

$A \subset C$

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Problem 7.1.7. Let $A = \{2,4,6,10\}$, $B = \{2,10\}$, and $C = \{2,4,6,10\}$. Decide whether the following are true or false.

$A \subseteq C$ and $C \subseteq A$

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Problem 7.1.8. Let $A = \{2,4,6,10\}$, $B = \{2,10\}$, and $C = \{2,4,6,10\}$. Decide whether the following are true or false.

$A = C$

Subsets

- $A \subseteq A$ for any set A.
- $\emptyset \subseteq A$
- A set with k elements has 2^k subsets.

ex. $\{1,2\}$ has 2^2 subsets.

$\{1,2\}$, $\{1\}$, $\{2\}$, $\emptyset$.

Problem 7.1.9. Let $E = \{2,4,6\}$

How many subsets does E have? List them.